

Title: Investigating the relationship between the coefficient of downforce produced by a set of hyperbolic-shaped wings at varying wind speeds.

Research Question: *How does varying the dilation constant (k , dimensionless) in the hyperbolic equation $y = \frac{-1}{kx}$ (which alters the curvature of a modelled custom rear wing) affect the coefficient of downforce (C_D , dimensionless), when exposed to five different wind speeds of 2ms^{-1} , 3ms^{-1} , 4ms^{-1} , 5ms^{-1} and 6ms^{-1} , measured experimentally using a force balance method?*

Subject: Physics

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Glossary

Custom wing: The Formula One (F1) rear wing used in this investigation was modelled as a custom hyperbolic shape mathematically using the curve $y = \frac{-1}{kx}$, where k is some positive integer from $k=1$ to $k=8$. It was then extruded across a fixed span (mm) and was chosen to model the theoretical downforce behaviour of an F1 car. This can be seen in Figure 0.

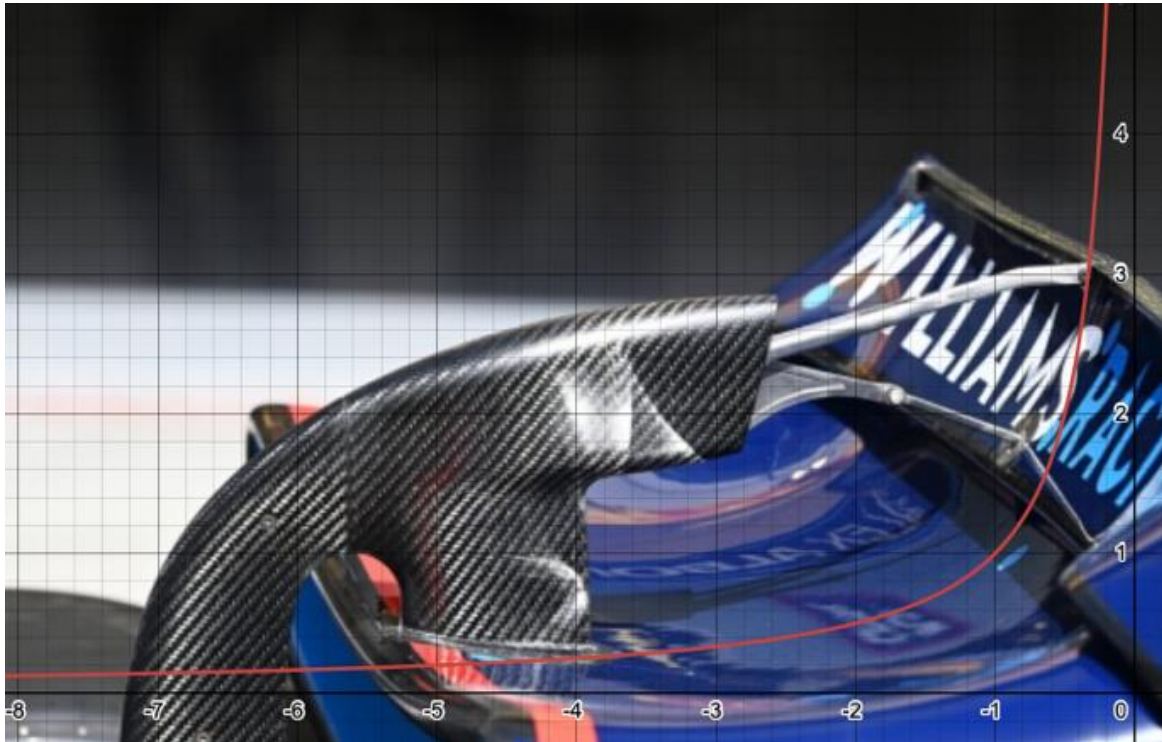


Figure 0. A $y = \frac{-1}{1x}$ graph plotted over a rear wing of a Formula One car, illustrating how the dilation factor works.

Downforce: The downforce force generated on a vehicle due to its aerodynamic features (*Continental AutoSports Ferrari, 2022*).

Leading face: The face of the custom aerofoil based on the hyperbolic curve that first interacts with the wind from the source.

Arc length: The length of the front curve (mm) of the custom aerofoil.

Dilation of wing: The k value, which determines the dilation of the custom wing, is shaped from the set of curves $y = \frac{-1}{kx}$.

Laminar flow: Fluid particles follow steady and smooth streamlines with minimal movement of particles between adjacent layers and is characterised by a low Reynolds number due to the viscous forces being much larger than the velocity.

Turbulent flow: Chaotic variations in the direction and magnitude of fluid particle velocity and amplitude of pressure. It is characterised by a high Reynolds number as the velocity is much larger than the viscous damping of the fluid.

Defining Variables

Symbol	Meaning
F_D	Downforce generated (N)
m	Mass produced (g)
ρ	Density of air (kgm^{-3})
v	Velocity of wind / wind speed (ms^{-1})
A	Surface Area of the custom wing (m^2)
C_D	Coefficient of Downforce (dimensionless)
S	Inner Arc Length of custom wing (mm)
w	Width of custom wing (mm)
k	Dilation of hyperbolic function (dimensionless)
g	Acceleration due to gravity (ms^{-2})
μ	Kinematic viscosity of the air (m^2s^{-1})

Table 0. Symbol of variables that will be used alongside their meaning.

1.0: Introduction

Formula One (F1) is the pinnacle of automotive engineering, pushing the limits of machinery and human skill. For years, the rear wing has been a crucial component, as it generates downforce to balance the car. Downforce is important for an F1 car as it allows turns to be taken at a higher speed and provides additional friction. However, excess downforce will cause the car to be slow on the straights due to the increase in linear air friction.

The key concepts underlying this phenomenon include the Lift Equation, Newton's Laws of Motion and the role of Reynolds number. These concepts allow for an understanding of aerodynamic behaviours such as airflow and downforce.

The primary objective of this report is to determine the relationship between the coefficient of downforce with the dilation of the wing at different wind speeds. This will be done by using a set of modelled Rear wings with different dilations, which will be 3D printed. By using a force balance method it allows for the direct measurement of downforce, which can be used to solve for the coefficient of downforce.

This topic was chosen due to my long-standing interest in F1. I have followed the sport closely, where I have seen the evolution of the rear wing. Furthermore, I have seen many times how a driver is unable to carry on due to a broken rear wing caused by the loss of downforce, which intrigued me to understand the generation of downforce. Hence, I aim to explore how the dilation of the rear wing and different wind speeds impact the coefficient of downforce.

2.0: Rear Wings

The first rear wing used in F1 was in 1968, which marked a revolutionary turn in the development of F1 cars, forcing teams to focus on aerodynamics, which at the time was new. It started through the use of inverted aerofoils, as traditional aerofoils are used to generate lift, but when inverted, they produce downforce. Teams like Ferrari and Lotus tested inverted aerofoils to force the car into the ground, hoping to generate more traction, seen in Figure 1.

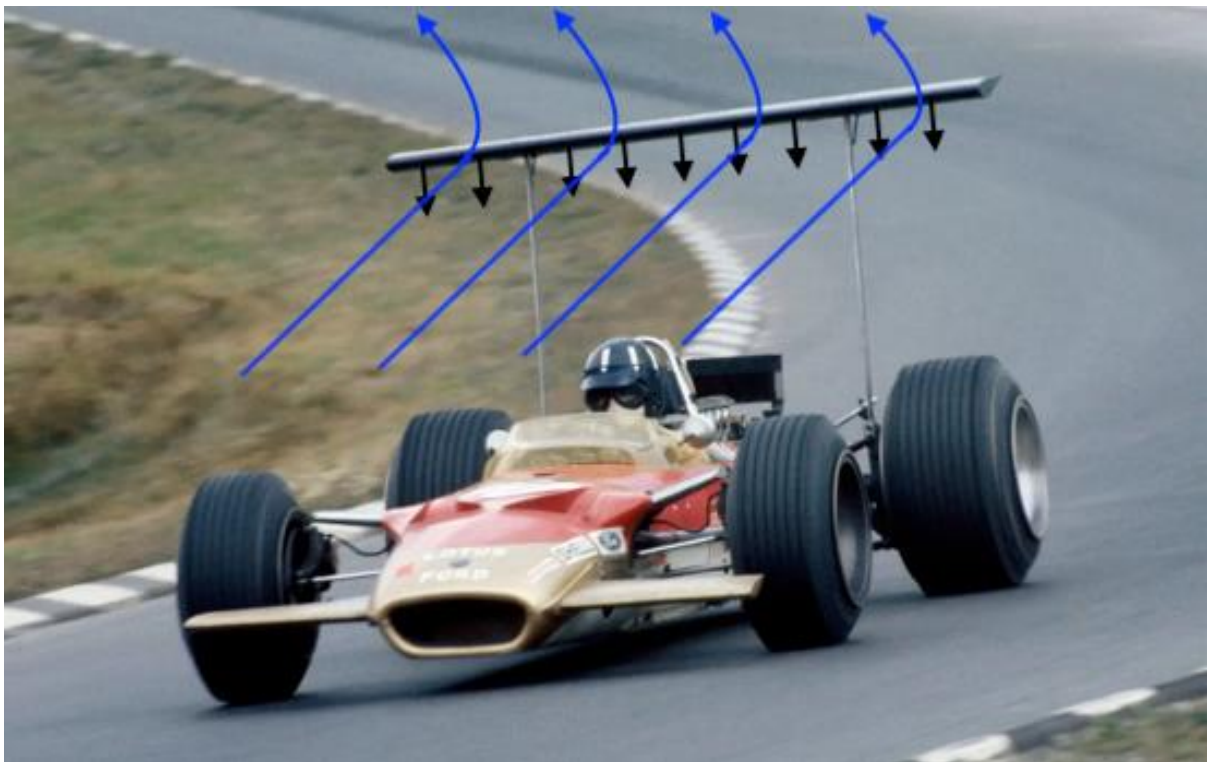


Figure 1. Ferrari testing an inverted aerofoil as a rear wing for the first time in 1968 (Phipps, n.d.), annotated with black lines to show the net force acting on the rear wing and blue lines to show the air flow.

Downforce is generated by manipulating airflow over a surface, creating a pressure difference. This pressure difference forces air to flow faster over its upper surface than its lower surface. This results in a high-pressure zone above the wing and a low-pressure one below the wing, resulting in a net downward force.

F1 rear wings have evolved every year, while early wings were simplistic, modern rear wings are highly optimised for specific tracks. This allows teams to adjust elements

such as wing angle, camber and dilation to balance downforce and drag, allowing them to be more suited to the demands of specific tracks. For example, when racing in Monaco, teams run a high-downforce setup due to the tight and low-speed corners. However, in Monza, Italy, teams run a low downforce setup due to the lack of turns, and it being a high-speed track, extra downforce will cause the car to be slow on the straights. Even within the same race weekend, teams will test different rear wing setups to find what is most suitable to the track and wind conditions of the day, as seen in Figure 2.

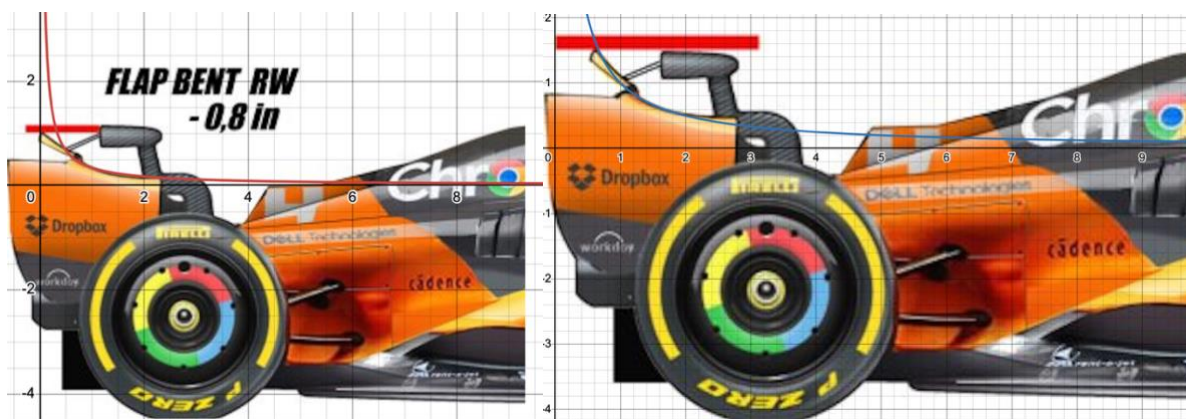


Figure 2. The change in dilation in the rear wing of a McLaren at different points of the race, plotted with the equation $y = \frac{-1}{kx}$. The left side has the equation $y = \frac{-1}{1x}$, and the right side has the equation $y = \frac{-1}{2x}$ (Filisetti, 2024).

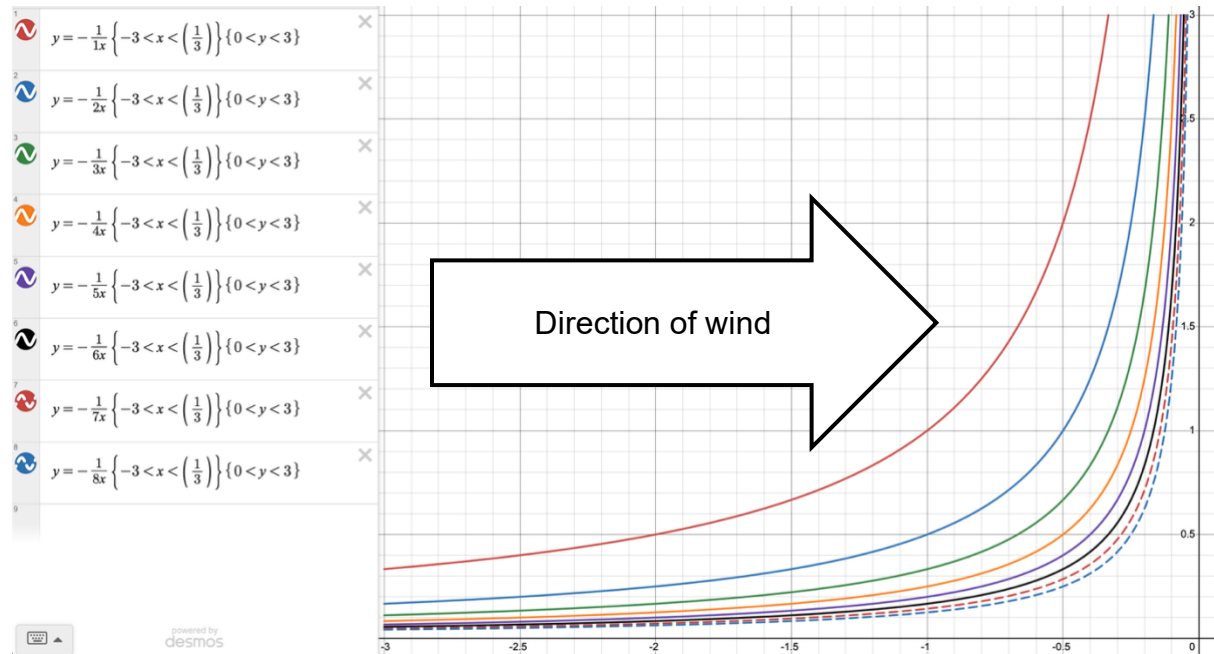
The use of wind tunnel testing has allowed engineers to experiment with new shapes, one focus being the dilation of the rear wing.

This experiment will mimic this development by testing a series of custom-designed wings shaped as a hyperbolic curve using the equation:

$$y = \frac{-1}{kx}$$

This mathematical representation allows for a controlled variation of dilation using the k value, allowing me to investigate the relationship between the dilation factor and the coefficient of downforce at different airspeeds.

I have plotted the graphs on Desmos in Graph 1 to illustrate how the dilations change alongside an angled view of the wings when the dilation factor is 1, 3, 5, and 7 in Figure 3 to illustrate the changes.



Graph 1. All the dilation factors of the rear wing that will be tested are plotted on Desmos and annotated with the direction of wind.

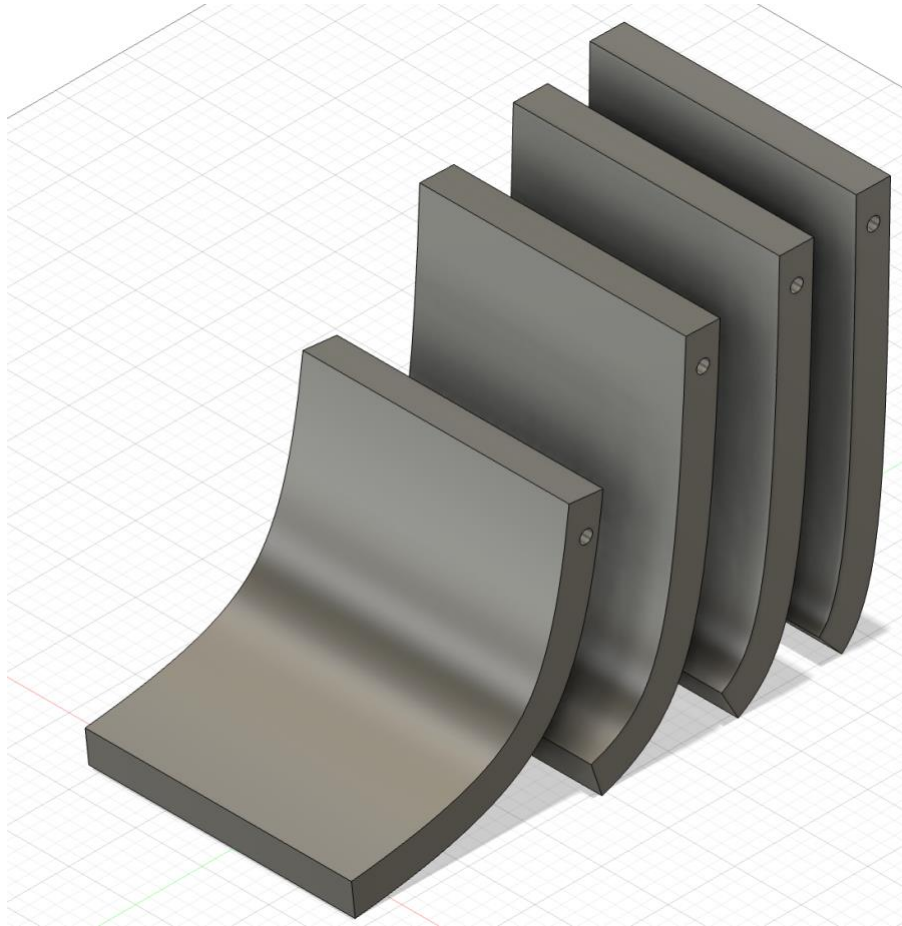


Figure 3. Angled view of the 3D wing when the dilation factor is 1, 3, 5 and 7 (Own Diagram).

3.0: Research Question

How does varying the dilation constant (k , dimensionless) in the hyperbolic equation $y = \frac{-1}{kx}$ (which alters the curvature of a modelled custom rear wing) affect the coefficient of downforce (C_D , dimensionless), when exposed to five different wind speeds of 2ms^{-1} , 3ms^{-1} , 4ms^{-1} , 5ms^{-1} and 6ms^{-1} measured experimentally using a force balance method (Hall, 2021)?

From this, I will be able to determine a relationship between the dilation factor at different wind speeds.

3.1: Design Justification

I have chosen not to use a conventional aerofoil as I am interested in knowing how downforce is generated on an F1 car varies through different designs of the rear wing. As teams experiment with different designs, it seems that a hyperbolic equation best fits how a rear wing is shaped.

I will use a force balance method because it will measure the net aerodynamic force (downforce) acting on the modelled wing by detecting the force exerted onto a balance. This allows for direct and quantitative measurements of downforce without requiring a wind tunnel, which I do not have access to.

There are no relevant safety or ethical considerations. Polylactic Acid (PLA) plastic was used as it is biodegradable, resulting in no environmental issues.

3.2: Variables

3.2.1: Independent Variable

The dilation of the Rear wing is the independent variable, which is defined by the dilation, k , in the hyperbolic equation $y = \frac{-1}{kx}$.

3.2.2: Dependent Variable

The coefficient of downforce is the dependent variable as it is calculated from the mass reading from the scale ($\pm 0.05\text{g}$), which is then applied in the derived formula.

3.2.3: Controlled Variable

- Windspeed: controlled at 2ms^{-1} , 3ms^{-1} , 4ms^{-1} , 5ms^{-1} and 6ms^{-1} by using the same wind source and distance setup to ensure that all trials received the same type of airflow.
- Angle of attack: Using a fixed 3D printed stand allows the wing to be held at a consistent orientation so only the dilation factor is changed.
- Material and infill of the custom wing: This was controlled by printing all eight wings at the same time on the same printer, ensuring that all wings had a 5% infill. Printing them all on same printer removed any chances of different infill output between different printers.
- Distance of custom wing from wind source: This was controlled by measuring 0.2m from the wind source to the custom wing. Once done, the balance and stand were fixed at that position, ensuring that all wings received the same air flow characteristics.

4.0: Background Information and Formula

Development

4.1: Downforce Equation and Newton's Laws

Newton's Third Law of Motion states that for every action, there is an equal but opposite reaction. When air interacts with the surface of the custom wing, its direction and momentum change, mainly redirecting upward, following the curvature created by the dilation of the wing. According to Newton's Third Law, the upward change in the air's momentum results in an equal and opposite reaction, which is the force exerted downforce on the wing, or known as downforce. However, this combined with Newton's Second Law will allow for the downforce to be calculated from the scale reading. Newton's Second Law states that the net force on a body of constant mass is proportional to the body's acceleration ($F = ma$) and is in the same direction as the acceleration. This is the law that explains that downforce is generated through the acceleration of air. As the air passes over the custom wing, it is redirected upward, resulting in a change in velocity, therefore a change in acceleration. However, to result in acceleration, the wing must exert a force on the air. In this investigation, the downforce is measured as an increase in the mass on a scale and is converted into force using:

$$F = m \times g \quad - (1)$$

The experiment will give data in the form of the mass produced by each custom wing. I will have to convert it into Newtons to determine the downforce that it produces, as downforce is measured in Newtons. This can be seen in equation 2, where F_D is the downforce produced, and m is the mass reading on the scale.

$$F_D = m \times g \quad - (2)$$

This information is then used in the lift equation, which is adapted for this experiment by renaming the coefficient as the coefficient of downforce and setting the force as positive

to avoid negative values, as positive lift corresponds to downforce. This is shown in equation 3.

$$F_D = C_D \times \frac{1}{2} \rho \times v^2 \times A \quad - (3)$$

Combining equations 2 and 3, I get:

$$m \times g = C_D \times \frac{1}{2} \rho \times v^2 \times A$$

$$C_D = \frac{2 \times m \times g}{\rho \times v^2 \times A} \quad - (4)$$

This formula will be used to calculate the experimental coefficient of downforce for each dilation of the custom wing in relation to different wind speeds.

4.2: Reynolds Number

To assess whether equation 4 is applicable under experimental conditions, the Reynolds number was calculated using the formula:

$$Re = \frac{\rho \times v \times S}{\mu} \quad - (5)$$

Where ρ is 1.225 kgm^{-3} , S is 0.135m , which is the arc length of the wing and μ is 1.470×10^{-5} at 15°C . The Reynolds number assesses whether the air flow over the wing is likely to be laminar or turbulent. This has an impact on the accuracy of my investigation, as I need to ensure that the wing faces laminar air flow. This is because an F1 car ideally experiences laminar airflow for its aerodynamic part to function effectively, such as the rear wing (Mafi, 2007). This suggests that the model I am using is usable to investigate the production of downforce by the rear wing.

Below is a sample calculation of the Reynolds number for the custom wing lap with a dilatation factor of 1 when facing wind at a speed of 3ms^{-1} .

$$Re = \frac{1.225 \times 3 \times 0.1327}{1.470 \times 10^{-5}}$$

$$Re = 33,175$$

The same process was used to find the Reynolds number for each custom wing at different wind speeds, which can be seen in Table 1.

Custom Wing Dilation k	Reynolds number at the wind speed of 2ms^{-1} Re_l	Reynolds number at the wind speed of 3ms^{-1} Re_l	Reynolds number at the wind speed of 4ms^{-1} Re_m	Reynolds number at the wind speed of 5ms^{-1} Re_h	Reynolds number at the wind speed of 6ms^{-1} Re_l
1	22,120	33,180	44,240	55,300	66,360
2	22,237	33,355	44,473	55,592	66,710
3	22,412	33,618	44,823	56,029	67,235
4	22,587	33,880	45,173	56,467	67,760
5	22,780	34,170	45,560	56,950	68,340
6	22,938	34,408	45,877	57,346	68,815
7	23,093	34,640	46,187	57,733	69,280
8	23,230	34,845	46,460	58,075	69,690

Table 1. The Reynolds number for each custom wing at different wind speeds.

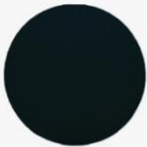
OPEN CHANNEL FLOW	PIPE FLOW	EXTERNAL FLOW OVER CYLINDER/ SPHERE	FLOW OVER FLAT SURFACE
			
Laminar $Re < 500$	Laminar $Re < 2,300$	Laminar $Re < 200,000$	Laminar $Re < 500,000$
Turbulent $Re > 1,000$	Turbulent $Re > 4,000$	Turbulent $Re > 200,000$	Turbulent $Re > 500,000$

Figure 4. Range of Reynolds number for different types of air flow (Thermal Book, 2016).

From my calculations, I found the range of Reynolds numbers for my investigation to be from 22,120 to 69,690. This is in the range for laminar external flow as seen in the section of external flow over a cylinder or sphere, according to Figure 4.

5.0: Experiment

5.1: Construction of Base

To support the custom wing during the experiment, I designed, and 3D printed a simple stand. This stand was placed on the precision scale while holding the custom wing, ensuring that all the vertical force resulting from the airflow would interfere with the custom wing.

The stand was 3D printed using PLA plastic and was designed in Fusion 360 to ensure that the connecting hole, in which a connector is placed to connect the stand to the wing, is the same. This design allows a pivoting mechanism, enabling the custom wing to rotate slightly under airflow, mimicking how the rear wing works when experiencing load, ensuring accurate data. The 3D model can be seen below alongside the dimensions.

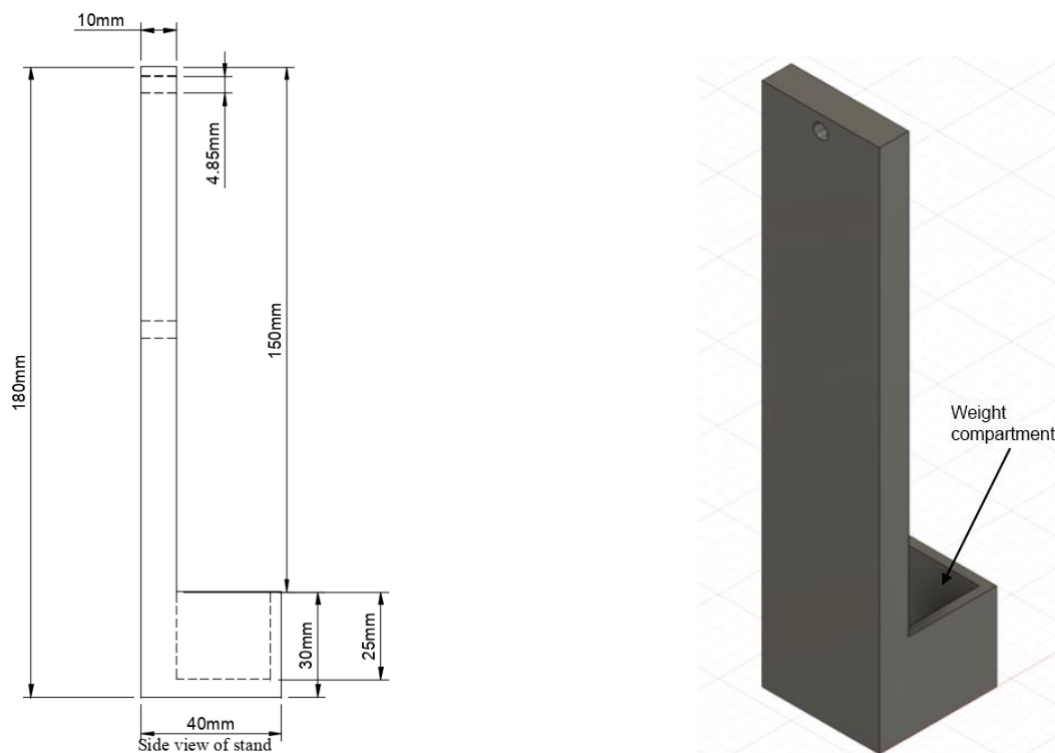


Figure 5. Side view of the stand with dimensions (left side) and the 3D view of the stand (right side) (Own Diagram).

To keep the stand stationary during testing, a weight compartment was built to its side, allowing small masses to be added to prevent the stand from shifting caused by the wind source's airflow.

5.2: Construction of the Rear Wing

The custom wing was 3D printed, allowing greater accuracy of measurements and ensuring all wings were made to the same standards. To ensure accuracy and similar conditions to all wings, the hyperbolic curve was written in TextEdit as Python code, which was imported into Fusion 360. The sample code is shown in Figure 5, with comments for when k was 3.

```
import adsk.core, adsk.fusion, traceback
import math

def run(context):
    ui = None
    try:
        app = adsk.core.Application.get()
        ui = app.userInterface
        design = app.activeProduct

        rootComp = design.rootComponent
        sketches = rootComp.sketches
        xzPlane = rootComp.xZConstructionPlane # Right face (XZ plane)
        sketch = sketches.add(xzPlane)

        points = adsk.core.ObjectCollection.create()

        scale_factor = 0.07 # shrink to make inner arc length = 135mm

        # x from -3 to -1/3
        steps = 300 # Number of steps for smoothness
        for i in range(steps + 1):
            x = -3 + (2.666 * i / steps) # from -3 to -1/3

            # Change the equation to  $y = -1/(3x)$ 
            y = -1 / (3 * x)

            x_scaled = x * scale_factor
            y_scaled = y * scale_factor

            points.add(adsk.core.Point3D.create(x_scaled, y_scaled, 0))

        sketch.sketchCurves.sketchFittedSplines.add(points)

    except:
        if ui:
            ui.messageBox('Failed:\n{}'.format(traceback.format_exc()))
```

Figure 6. Sample code used to input the hyperbolic relationship into Fusion 360 to then create the custom wing (Own Diagram).

The custom rear wing was designed using a hyperbolic function, with the inner and outer curves created by duplicating the function and offsetting it by 12.85mm. Due to the different functions, the range of the inner curve was between 135mm and 137mm. However, the total surface area was used when calculating the coefficient of downforce, minimising the impact on accuracy. The curves were connected by straight lines and extruded 100mm to create a 3D wing. Mounting holes near the top of the wing allowed minimal rotational movement using a connector. The wings were 3D printed at a 5% infill, making them sturdy but light, minimising their impact on aerodynamic performance. A sample wing in Figure 6 represents an annotated side view of the wing when k is 3, and Figure 7 shows an angled view of the wing when k is 1.

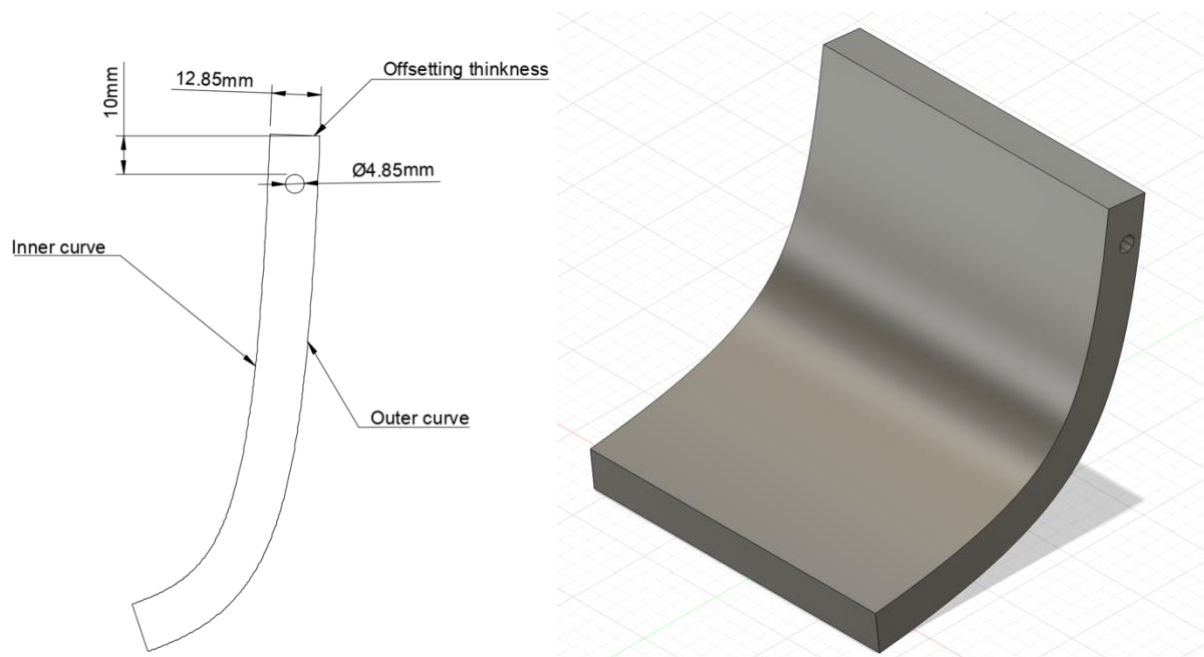


Figure 7. An annotated side view of the custom wing when k is 3 (right side) and an angled view of the custom wing when the dilation factor (k) is 1 (left side) (Own Diagram).

5.3: Materials

- Wind source with five different speeds
- 16 Lego connectors
- 1 electronic balance ($\pm 0.05\text{g}$)
- 1 retort stand
- 1 burette clamp
- 1 anemometer ($\pm 0.5\text{ms}^{-1}$)
- 1 stopwatch
- Access to a 3D printer with PLA
- Access to Fusion 360
- Access to a Python coding software

5.4: Methodology

1. Using an anemometer, find out the wind speed of the five different speed settings of the wind source. Ensure that the anemometer is 0.20m away from the wind source.
2. The apparatus set up is shown in Figure 9, where the wing is fitted in-between the stands. Both stands are then placed on the electronic balance. Set up the wind source on a retort stand while being clamped, adjust it so that the wind source is pointed to the centre of the wing, ensuring that the wind source is 0.20m away from the wing.

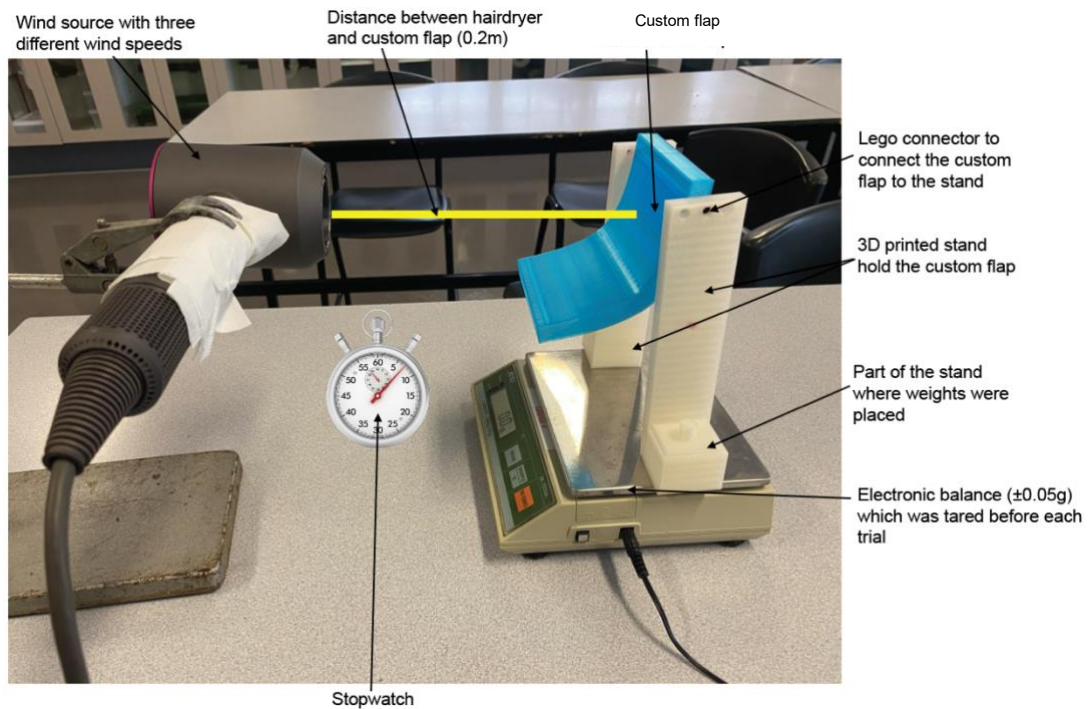


Figure 8. Model apparatus with comments (Own Diagram).

3. Turn on the wind source at its lowest/first speed.
4. Wait for 10 seconds, then read the mass on the scale.
5. Repeat five times.
6. Repeat this for the same wing but using the wind source's second, third, fourth and fifth speed.
7. Repeat above for all wings.

6.0: Analysis

6.1: Raw Data

Wind speed of $2 \pm 0.5 \text{ms}^{-1}$			
Custom Wing Dilation k	Trial 1 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 2 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 3 Mass Produced by custom wing m / g $\pm 0.05\text{g}$
1	12.4	12.2	12.4
2	9.8	9.5	9.9
3	9.2	9.3	9.6
4	7.9	8.1	8.3
5	6.8	6.9	6.7
6	7.6	7.7	7.4
7	8.7	8.8	8.7
8	10.0	9.5	9.7

Table 2. Raw data with mass produced in grams by the different dilated custom wings at a wind speed of 2ms^{-1} .

Wind speed of $3 \pm 0.5 \text{ms}^{-1}$			
Custom Wing Dilation k	Trial 1 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 2 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 3 Mass Produced by custom wing m / g $\pm 0.05\text{g}$
1	26.1	25.8	25.9
2	20.1	19.9	19.8
3	18.9	19.1	19.3
4	17.4	17.1	17.2
5	13.5	13.9	13.7
6	15.2	15.1	15.2
7	17.5	17.6	17.9
8	19.4	19.4	19.0

Table 3. Raw data with mass produced in grams by the different dilated custom wings at a wind speed of 3ms^{-1} .

Wind speed of $4 \pm 0.5 \text{ms}^{-1}$			
Custom Wing Dilation <i>k</i>	Trial 1 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 2 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 3 Mass Produced by custom wing m / g $\pm 0.05\text{g}$
1	43.1	43.2	42.9
2	32.2	31.6	32.4
3	30.1	30.0	30.0
4	25.1	24.8	24.8
5	21.7	22.1	21.9
6	24.3	24.4	24.3
7	28.1	27.8	28.0
8	27.2	26.4	26.5

Table 4. Raw data with mass produced in grams by the different dilated custom wings at a wind speed of 4ms^{-1} .

Wind speed of $5 \pm 0.5 \text{ms}^{-1}$			
Custom Wing Dilation <i>k</i>	Trial 1 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 2 Mass Produced by custom wing m / g $\pm 0.05\text{g}$	Trial 3 Mass Produced by custom wing m / g $\pm 0.05\text{g}$
1	55.3	55.5	55.1
2	39.8	40.3	40.2
3	39.0	38.8	38.8
4	31.4	31.7	31.8
5	28.2	28.5	28.6
6	31.4	30.9	31.0
7	36.0	36.3	36.5
8	38.8	38.5	38.5

Table 5. Raw data with mass produced in grams by the different dilated custom wings at a wind speed of 5ms^{-1} .

Wind speed of $6 \pm 0.5 \text{ ms}^{-1}$			
Custom Wing Dilation k	Trial 1 Mass Produced by custom wing m / g $\pm 0.05 \text{ g}$	Trial 2 Mass Produced by custom wing m / g $\pm 0.05 \text{ g}$	Trial 3 Mass Produced by custom wing m / g $\pm 0.05 \text{ g}$
1	74.8	74.5	74.6
2	54.1	53.9	54.0
3	51.2	51.4	51.7
4	48.8	48.4	48.8
5	37.0	37.0	36.8
6	40.3	40.1	40.2
7	46.9	46.3	47.0
8	50.0	50.1	50.1

Table 6. Raw data with mass produced in grams by the different dilated custom wings at a wind speed of 6 ms^{-1} .

6.2: Data Processing

Using the raw data collected in Tables 2 to 6, I can develop a set of data for the average of the trials, as this will limit the impacts of random error for the mass produced by each custom wing at different wind speeds which will be called $m_{average}$. I will use the data of when the k was 3 and the wind speed was 3 ms^{-1} (highlighted in Table 3) as a sample calculation; the other trials will be processed automatically in Excel.

$$m_{average} = \frac{(18.9 \pm 0.05) + (19.1 \pm 0.05) + (19.3 \pm 0.05)}{(3 \pm 0)}$$

$$m_{average} = 19.1 \pm 0.05 \text{ g}$$

$$m_{average} = (19.1 \pm 0.05) \times 10^{-3} \text{ kg}$$

Custom Wing Dilation k	Inner Arc Length of Custom Wing S / mm $\pm 0.5\%$	Width of Custom Wing w / mm $\pm 0.5\%$
1	132.72	100.00
2	133.42	100.00
3	134.47	100.00
4	135.52	100.00
5	136.68	100.00
6	137.63	100.00
7	138.56	100.00
8	139.38	100.00

Table 7. Dimensions of each custom wing. The lengths have an uncertainty of $\pm 0.5\%$ as it is the standard uncertainty of a 3D printer (Formlabs, 2020).

Using these dimensions alongside its uncertainties, I will calculate the surface area (m^2), A , of the custom wing when k was 3. However, both the inner arc length and width will have to be divided by 1000 to be in meters, which is the standard unit for distance.

$$A = \frac{S}{1000} \times \frac{w}{1000}$$

$$A = \frac{134.47 \pm 0.5\%}{1000} \times \frac{100.00 \pm 0.5\%}{1000}$$

$$A = (0.13447 \pm 0.5\%) \times (0.10000 \pm 0.5\%)$$

$$A = 0.013447 \pm 1\%$$

$$A = 0.0134 \pm 0.0001 \text{ m}^2$$

6.3: Calculating Coefficient of Downforce

The coefficient of downforce, C_D , will be calculated for each trial. This will be done by using the formula deduced in the introduction, $C_D = \frac{2 \times m \times g}{\rho \times v^2 \times A}$, where $m_{average}$ is the average mass produced by the custom wing (kg), ρ is the density of air, which is assumed to be 1.225 kgm^{-3} at 15°C at standard pressure (Engineering Toolbox, 2003), v is the wind speed (ms^{-1}), A is the surface area of the wing (m^2) and g is the acceleration due to gravity which is

9.81 ms⁻² (Nasa, 2023). I will use the data from when k was 3 at the wind speed of 3ms⁻¹ as a sample calculation.

$$C_D = \frac{2 \times m \times g}{\rho \times v^2 \times A}$$

$$C_D = \frac{(2 \pm 0) \times (0.0191 \pm 0.00005) \times (9.81 \pm 0)}{(1.225 \pm 0) \times (3 \pm 0.5)^2 \times (0.013447 \pm 0.00013447)}$$

$$C_D = \frac{(2 \pm 0\%) \times (0.0191 \pm 0.261\%) \times (9.81 \pm 0\%)}{(1.225 \pm 0\%) \times (3 \pm 16.667\%)^2 \times (0.013447 \pm 1\%)}$$

$$C_D = \frac{(0.374742 \pm 0.261\%)}{(0.148253175 \pm 34.334\%)}$$

$$C_D = 2.5277165228 \pm 34.595\%$$

$$C_D = 2.528 \pm 34.595\%$$

$$C_D = 2.528 \pm 0.875$$

$$C_D = 2.5 \pm 0.8$$

The same method will be used to calculate all the data values with their uncertainties using an Excel spreadsheet. The final values can be seen in Tables 8 to 12.

Wind speed of $2 \pm 0.5\text{ms}^{-1}$	
Custom Wing Dilation k	Coefficient of Downforce C_D
1	3.7 ± 1.9
2	2.9 ± 1.5
3	2.8 ± 1.4
4	2.4 ± 1.2
5	2.0 ± 1.0
6	2.2 ± 1.1
7	2.5 ± 1.3
8	2.8 ± 1.4

Table 8. Custom Wing Dilation vs Coefficient of Downforce with absolute uncertainty when faced at a wind speed of 2ms⁻¹.

Wind speed of $3 \pm 0.5\text{ms}^{-1}$	
Custom Wing Dilation k	Coefficient of Downforce C_D
1	3.5 ± 1.2
2	2.7 ± 0.9
3	2.5 ± 0.9
4	2.3 ± 0.8
5	1.8 ± 0.6
6	2.0 ± 0.7
7	2.3 ± 0.8
8	2.5 ± 0.9

Table 9. Custom Wing Dilation vs Coefficient of Downforce with absolute uncertainty when faced at a wind speed of 3ms^{-1} .

Wind speed of $4 \pm 0.5\text{ms}^{-1}$	
Custom Wing Dilation k	Coefficient of Downforce C_D
1	3.2 ± 0.8
2	2.4 ± 0.6
3	2.2 ± 0.6
4	1.8 ± 0.5
5	1.6 ± 0.4
6	1.8 ± 0.5
7	2.0 ± 0.5
8	1.9 ± 0.5

Table 10. Custom Wing Dilation vs Coefficient of Downforce with absolute uncertainty when faced at a wind speed of 4ms^{-1} .

Wind speed of $5 \pm 0.5\text{ms}^{-1}$	
Custom Wing Dilation k	Coefficient of Downforce C_D
1	2.7 ± 0.6
2	1.9 ± 0.4
3	1.9 ± 0.4
4	1.5 ± 0.3
5	1.3 ± 0.3
6	1.4 ± 0.3
7	1.7 ± 0.4
8	1.8 ± 0.4

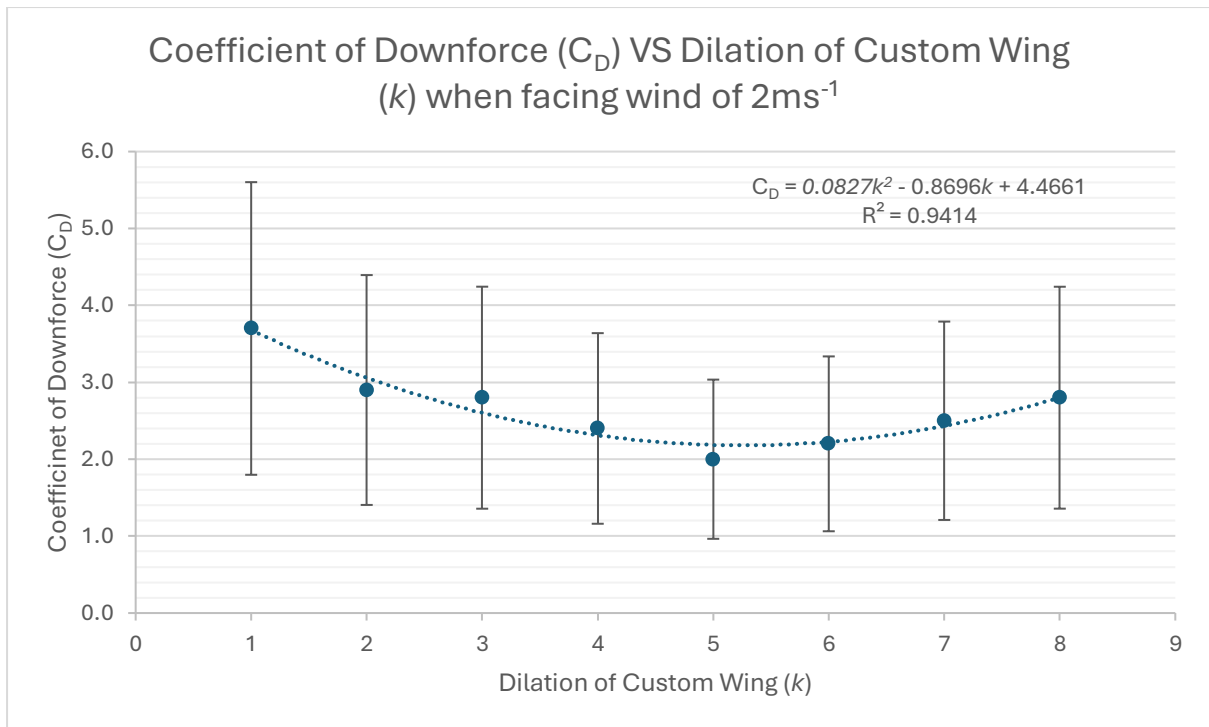
Table 11. Custom Wing Dilation vs Coefficient of Downforce with absolute uncertainty when faced at a wind speed of 5ms^{-1} .

Wind speed of $6 \pm 0.5 \text{ms}^{-1}$	
Custom Wing Dilation k	Coefficient of Downforce C_D
1	2.5 ± 0.4
2	1.8 ± 0.3
3	1.8 ± 0.3
4	1.6 ± 0.3
5	1.1 ± 0.2
6	1.3 ± 0.2
7	1.5 ± 0.3
8	1.6 ± 0.3

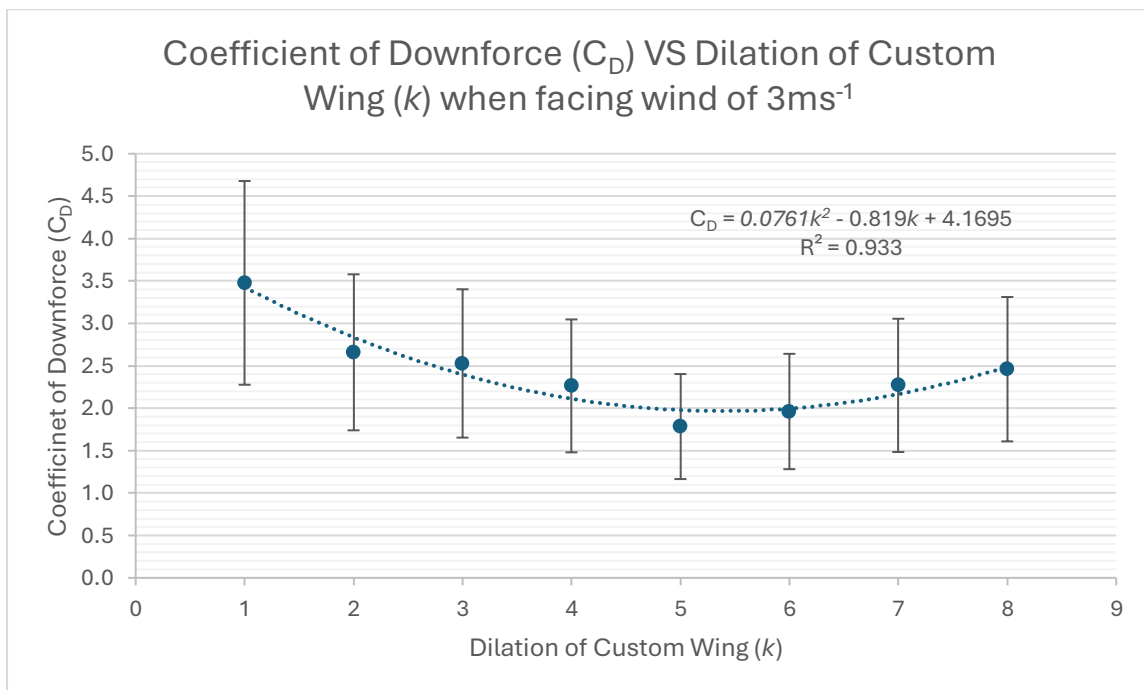
Table 12. Custom Wing Dilation vs Coefficient of Downforce with absolute uncertainty when faced at a wind speed of 6ms^{-1} .

The data tables suggest that there will be a minimum coefficient of downforce, hence I will try using a quadratic model to build a relationship between the coefficient of downforce (C_D) and the dilation of the custom wing (k), there is no physical reason for this decision.

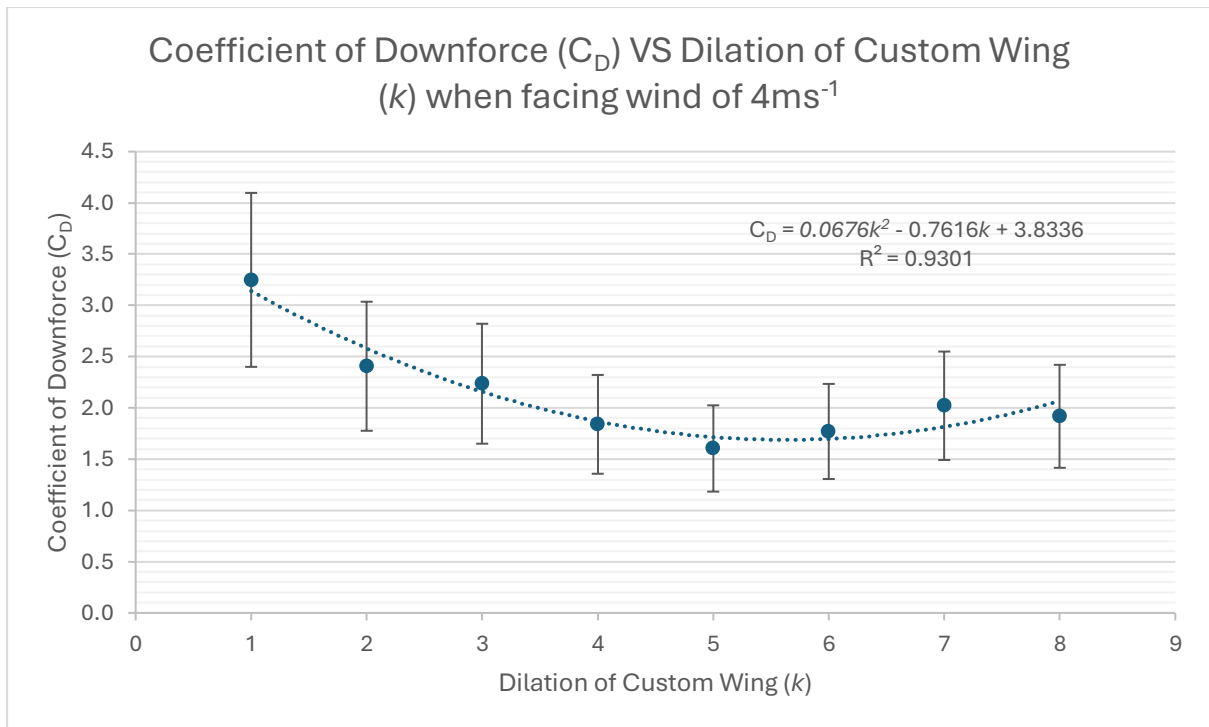
I then graphed these tables to get a graphical understanding of the relationship seen in Graphs 2 to 6.



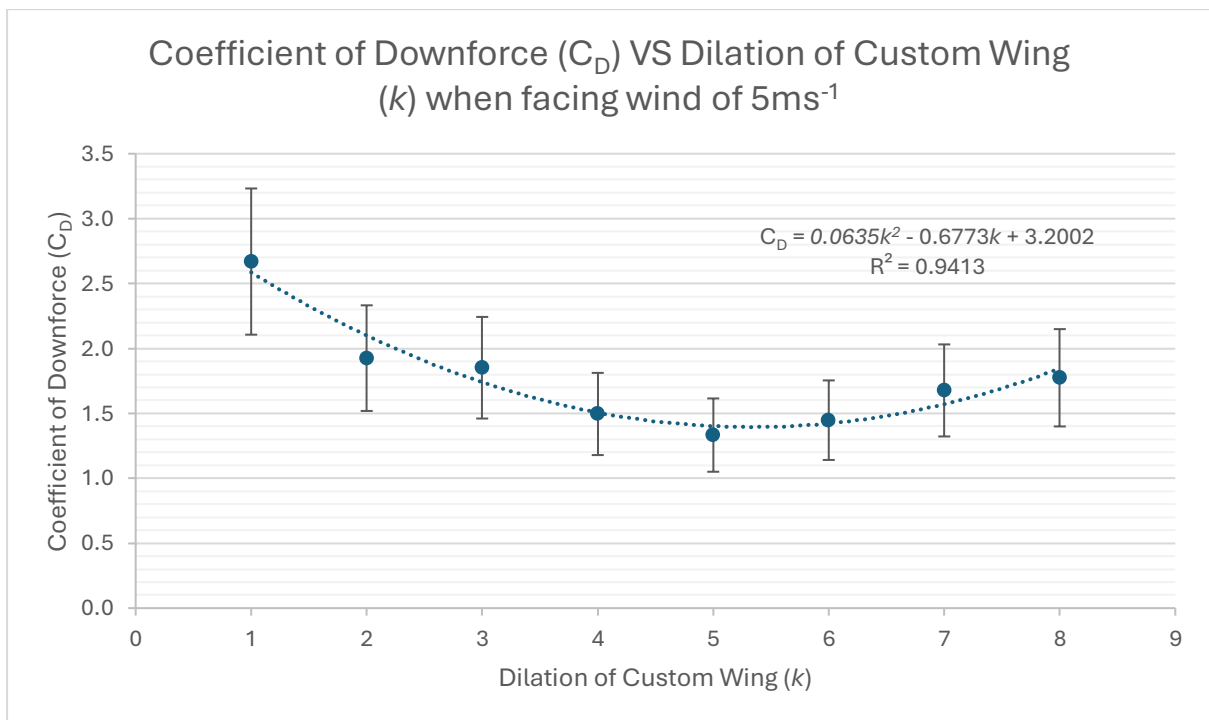
Graph 2. Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind of 2ms^{-1} .



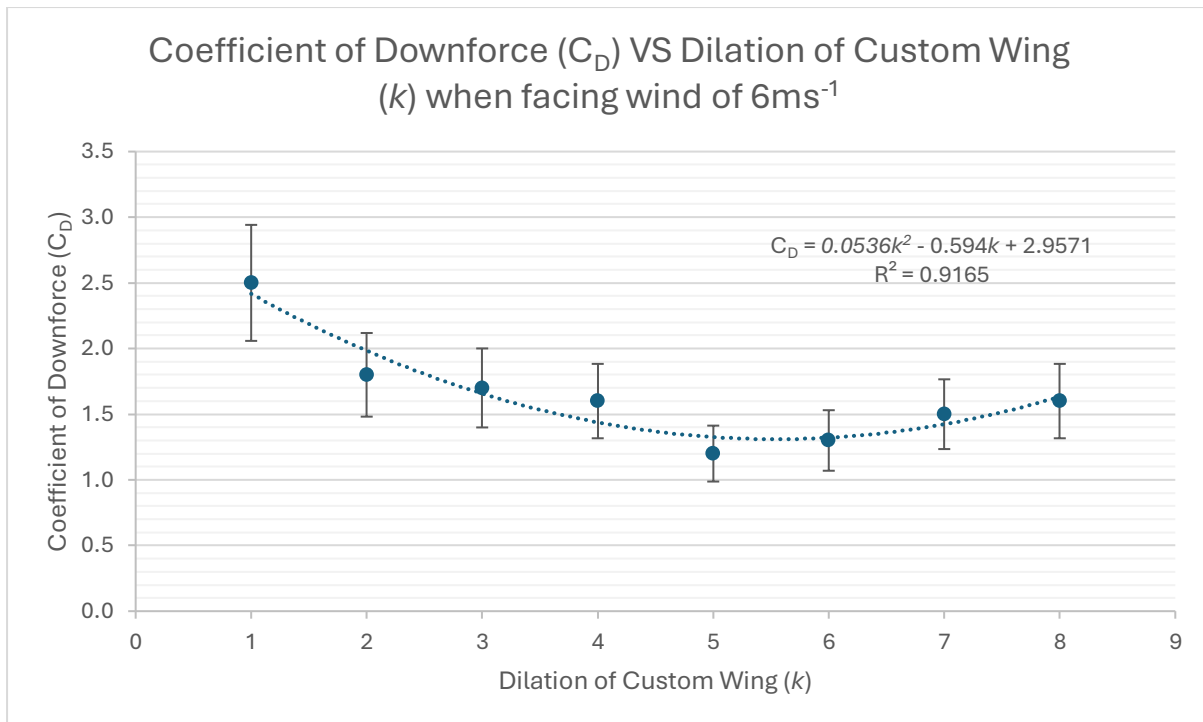
Graph 3. Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind of 3ms^{-1} .



Graph 4. Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind of 4ms^{-1} .



Graph 5. Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind of 5ms^{-1} .

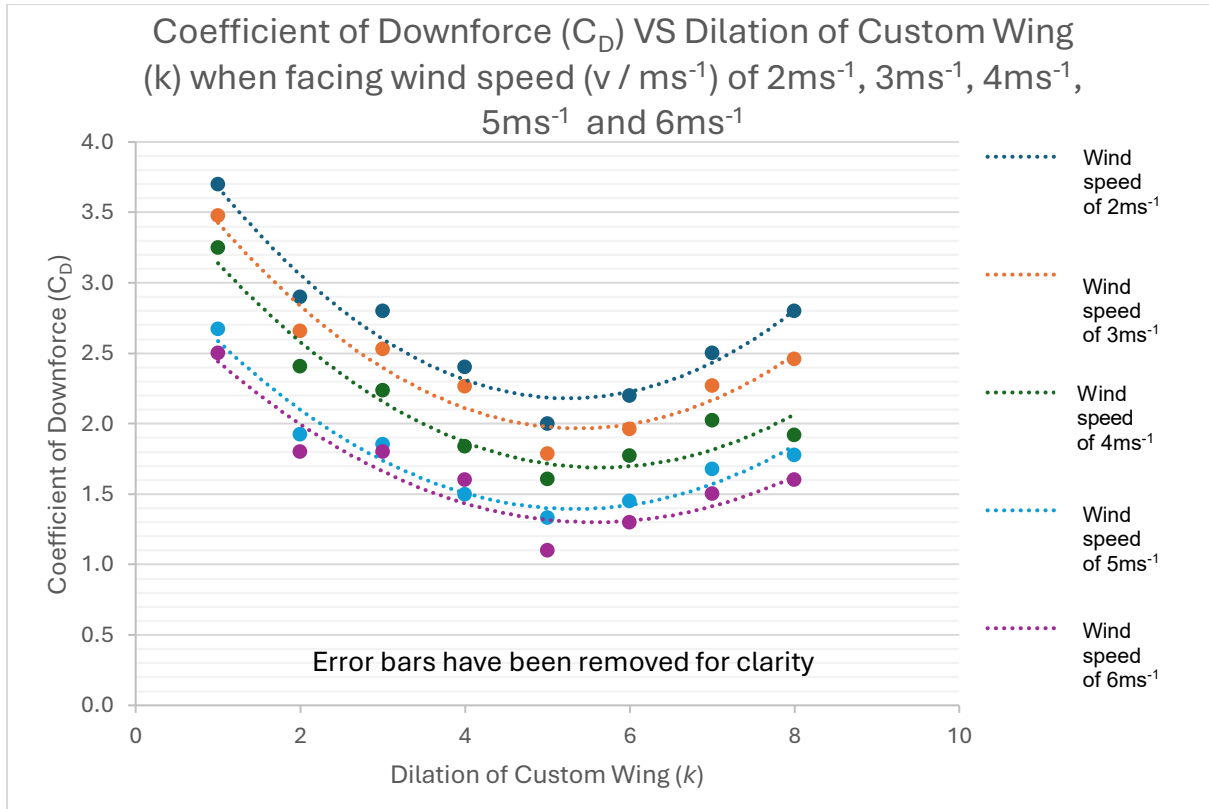


Graph 6. Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind of 6ms^{-1} .

All the graphs suggest a very strong relationship, as the range of the R^2 value is from 0.9165 to 0.9414. However, the size of the error bars decreased when the wind speed increased, as the uncertainty in the anemometer started to become limited. All graphs showcase a similar relationship; only the constant was different. This demonstrates how the trend is repeatable, as all graphs follow a similar curvature, with only the magnitude differing.

Overall, all five graphs suggest a very strong relationship, suggesting highly accurate data.

The five curves from Graphs 2 to 6 are plotted onto one graph to get a graphical representation of the trends at different wind speeds seen in Graph 7.

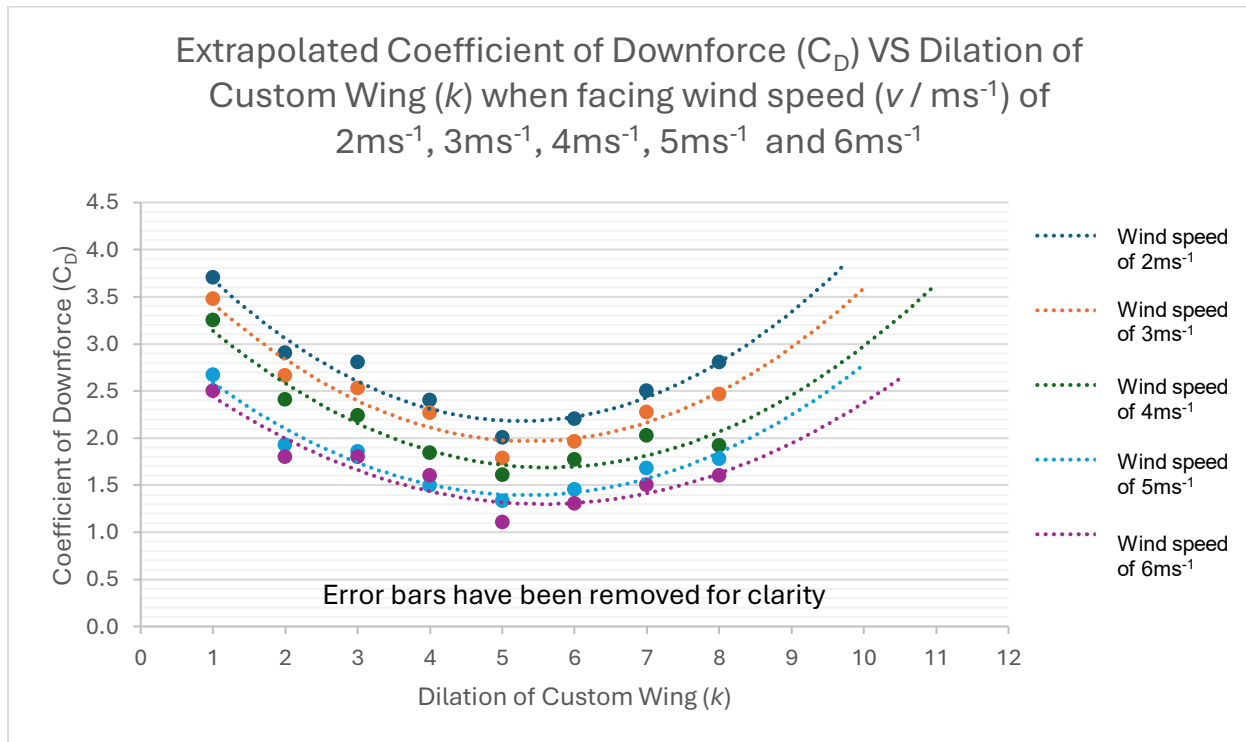


Graph 7. Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind speed (v / ms^{-1}) of 2ms^{-1} , 3ms^{-1} , 4ms^{-1} , 5ms^{-1} , and 6ms^{-1} .

It is observed from Graph 7 that regardless of the wind speed, the dilation of the custom wing and the coefficient of downforce follow a similar trend, all being parabolic-like and having the minimum coefficient of downforce when k is 5. After this, all curves start to increase, forming a consistent trend. The only difference is the magnitude of the coefficient of downforce. It is seen that the lowest wind speed produces the greatest magnitude of the coefficient of downforce. This can be explained as from equation 4. The velocity is in the denominator and is squared; hence, a lower velocity results in a lower denominator, increasing the value of the coefficient. All curves illustrate a repeatable trend, suggesting high reliability and predictability in the aerodynamic performance of the modelled rear wings.

To extend this investigation, I have extrapolated the relationship beyond the tested dilation range. This will allow theoretical analysis of extreme configurations, such as whether different factors can produce the same downforce coefficient, as shown in Graph 6. Despite the extrapolated values not being validated experimentally, they offer insight into the

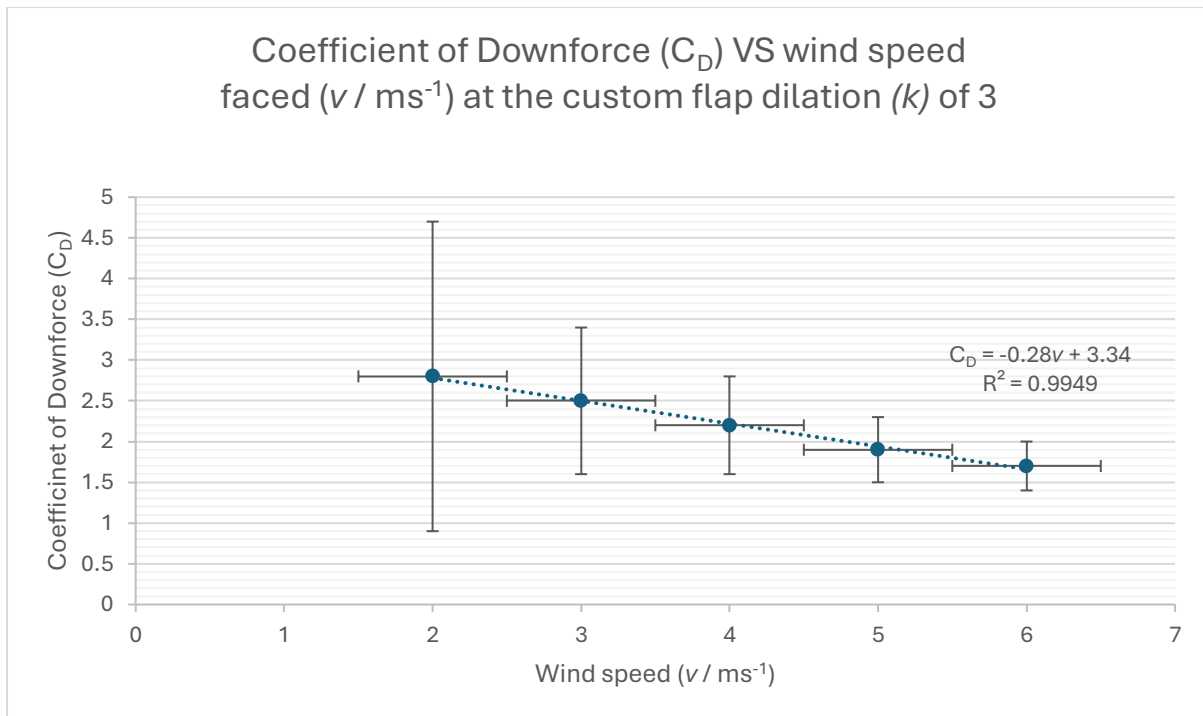
behaviour of the system outside the measured domain and highlight the potential aerodynamic trade-offs in wing design.



Graph 8. Extrapolated Coefficient of Downforce (C_D) VS Dilation of Custom Wing (k) when facing wind speed (v / ms^{-1}) of 2ms^{-1} , 3ms^{-1} , 4ms^{-1} , 5ms^{-1} and 6ms^{-1} .

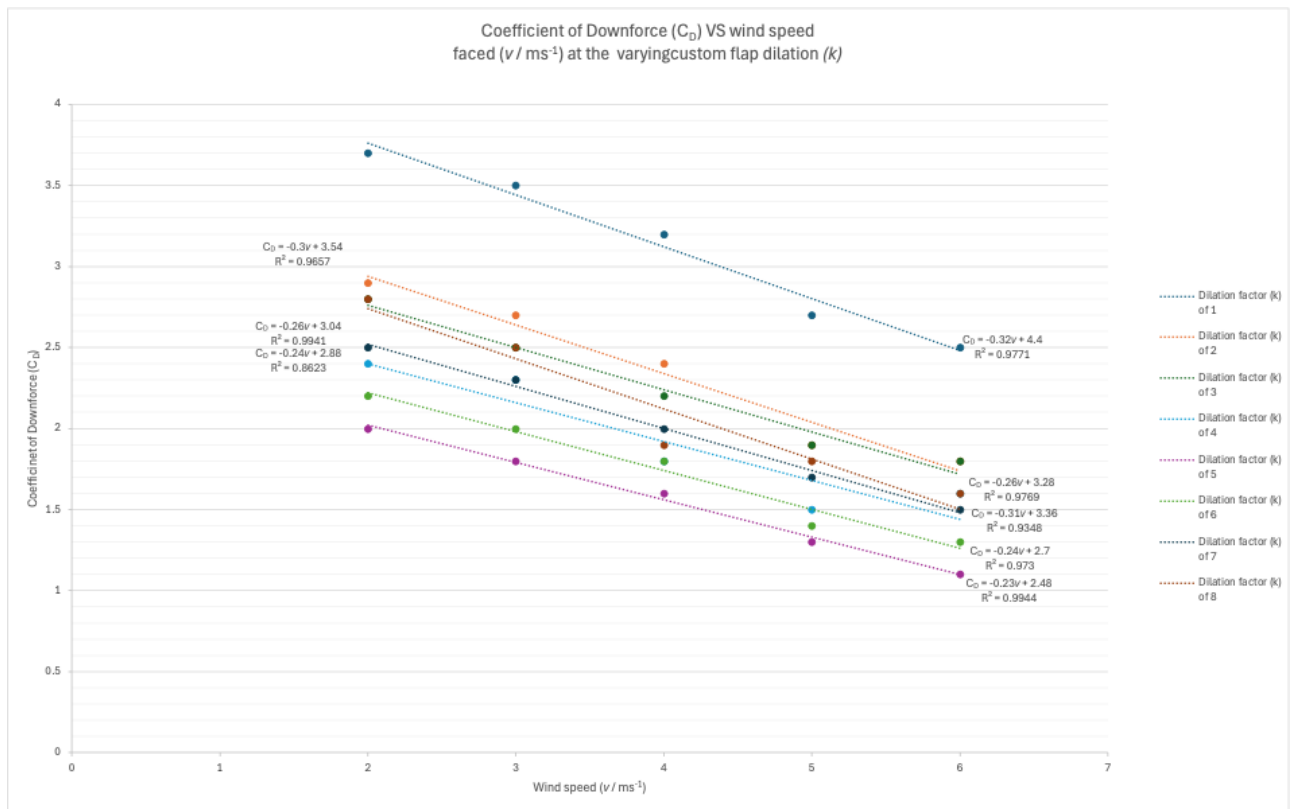
Graph 8 shows that at the wind speed of 5ms^{-1} , if the custom wing were k is around 10, it would produce the same downforce coefficient as when k is 1.

A graph will be plotted of the downforce coefficient values that I have obtained against the five wind speeds when k is 3. This will allow me to develop a relationship between the coefficient of downforce and wind speed. This can be seen in Graph 9.



Graph 9. Coefficient of Downforce (C_D) VS wind speed faced (v / ms^{-1}) at the custom wing dilation (k) of 3 with error bars.

Graph 9 shows a linear relationship between the coefficient of downforce and the wind speed when k is 3. An R^2 value of 0.9949 suggests a linear model is most applicable for this relationship. The error bars gradually decrease as the wind speed increases, showing how the uncertainty in the anemometer reduces at higher speeds. To verify if all the dilation factors follow a similar trend, I will plot all the dilation factors on Graph 8.



Graph 10. Coefficient of Downforce (C_D) VS wind speed faced (v / ms^{-1}) at the varying custom wing dilation of 3. The error bars were removed for clarity.

Graph 8 shows that all dilation factors follow a linear relationship when the downforce coefficient is plotted against the wind speed. Despite them all showing different equations of the relationship, they are all linear, creating a repeatable trend emphasised by the R^2 values above 0.973.

8.0: Conclusion

This investigation aimed to answer the research question: How does varying the dilation constant (k , dimensionless) in the hyperbolic equation $y = \frac{-1}{kx}$ (which alters the curvature of a modelled custom rear wing) affect the coefficient of downforce (C_D , dimensionless), when exposed to five different wind speeds of 2ms^{-1} , 3ms^{-1} , 4ms^{-1} , 5ms^{-1} and 6ms^{-1} , measured experimentally using a force balance method?

The results showed a consistent, parabolic-like relationship between the dilation factor and the coefficient of downforce across all the tested wind speeds. Although a higher-degree polynomial could have been used, a quadratic model was used to develop a relationship, as a parabolic trend was present in all data sets. It was a strong fit, emphasised by the R^2 values exceeding 0.91.

The quality of data was generally strong, as consistent and repeatable trends were observed across all wind speeds. However, noticeable variations were present in the form of large error bars, particularly in the lower factors of dilation. This variability is mostly caused by the inaccuracy of the anemometer, but this uncertainty has not significantly undermined the overall trend.

An interesting implication of this model is that, when extrapolated beyond the tested range, different dilation values could theoretically produce the same downforce coefficient. As seen in Graph 8, a k value of around 10 would produce the same coefficient of downforce as when k is 1.

A strength of this experiment was that it aligned with aerodynamic theories because the downforce coefficient became more stable at higher wind speeds across all dilation factors. This is because an increase in Reynolds number encourages the transition of the air flow from laminar to turbulent (White, 2016). Despite turbulent flow being more chaotic, it carries higher momentum near the surface, allowing it to adhere to the surface easily

(Anderson, 2016). This reduces flow separation and pressure drag (Munson et al, 2017), leading to more consistent and predictable downforce coefficients.

A theoretical limitation is related to scaling, as the model was small and operated at relatively low wind speeds, resulting in lower Reynolds numbers than those in real F1 applications (Azmi et al., 2017). Despite the lower Reynolds numbers, they are in the range of laminar flow, which is used by F1 teams during testing, which assumes ideal conditions (Deepak, 2024). This discrepancy affects how airflow behaves over the surface of the wing and limits the applicability of these results to full-scale race conditions. Nonetheless, within the scope of the experiment, the investigation successfully demonstrated a consistent and non-linear relationship between dilation and downforce as well as a linear relationship between downforce and wind speeds at a particular dilation factor (k).

To summarise, regardless of the wind speed, the dilation of the custom wing and the coefficient of downforce follow a similar trend, all being parabolic-like and having the minimum coefficient of downforce when k is 5. Beyond this point, all curves increase, forming a consistent trend, only differing in the magnitude of the coefficient of downforce.

9.0: Evaluation

Despite the strengths of this investigation, there are a few limitations that affect the validity and the generalisability of the results. An aspect impacting data accuracy is that the wind source produces turbulent air, possibly creating air that curves or circles around the top of the wing. This can generate greater force, resulting in higher mass produced, leading to a greater coefficient of downforce. This would consistently influence the results in the same direction, representing a systematic error. Since I did not have access to a wind tunnel, I could not ensure that the wind that interacts with the wing is constant. For future experiments, it is recommended to use a wind tunnel, as this would allow me to observe how the air diverges. A wind tunnel would ensure that the custom wing receives a consistent speed of air throughout, without circulation at the top of the wing, as seen in Figure 9. Additionally, a wind tunnel would allow testing of greater wind speeds, overcome the limitations of my current wind source and enable the measurement of drag force, which is beneficial for future investigations.

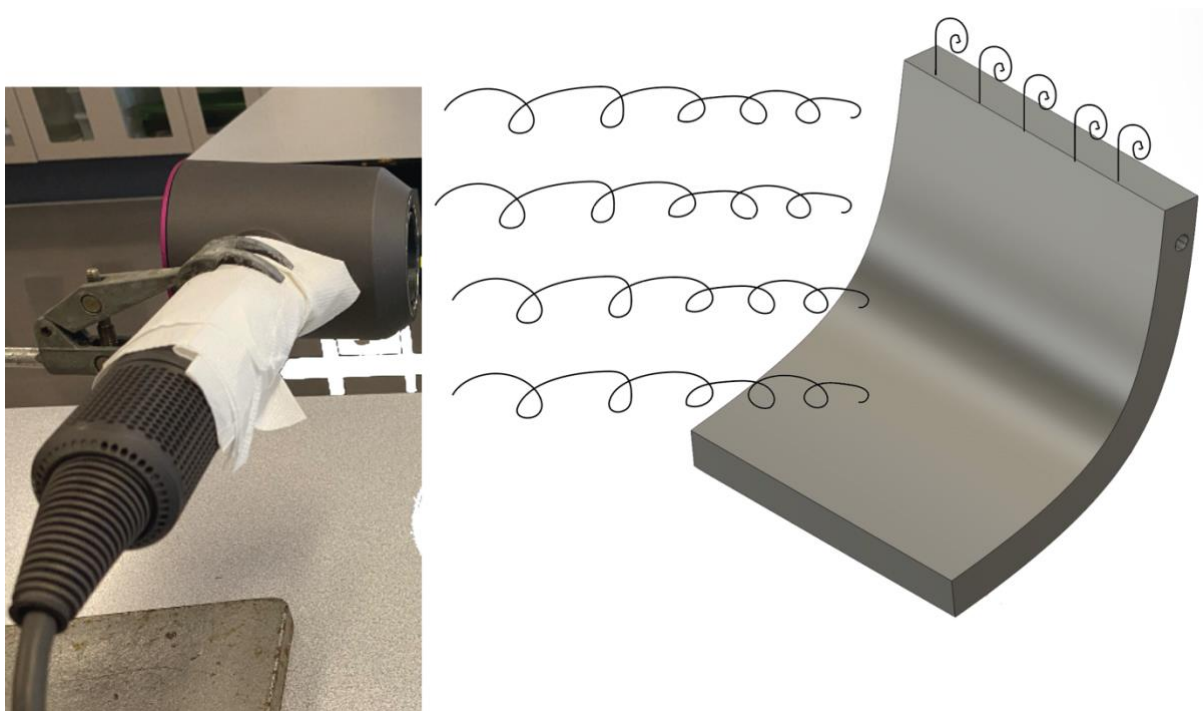


Figure 9. The curling of air at the custom rear wing is produced by a turbulent air source.

Through the calculation of the Reynolds number, it suggests that there is a splitting of air flow caused by the number of vertices at the lower dilation factors of the custom wing. A possible reason for this is that the modelled wings from k is 1 to k is 3 have a great degree of dilation, which forces the 3D printer to create more vertices to print the flap accurately. These vertices create vortices which create a low-pressure area under the flap, creating more downforce (Abdessalem Bouferrouk, 2014), resulting in the splitting of airflow and resulting in less stable downforce coefficients over a given dilation factor as seen in Figure 10. For future experiments, it would be beneficial to 3D print the custom wings with a more accurate 3D printer to ensure that the surfaces are smoother, not allowing any air to be trapped.

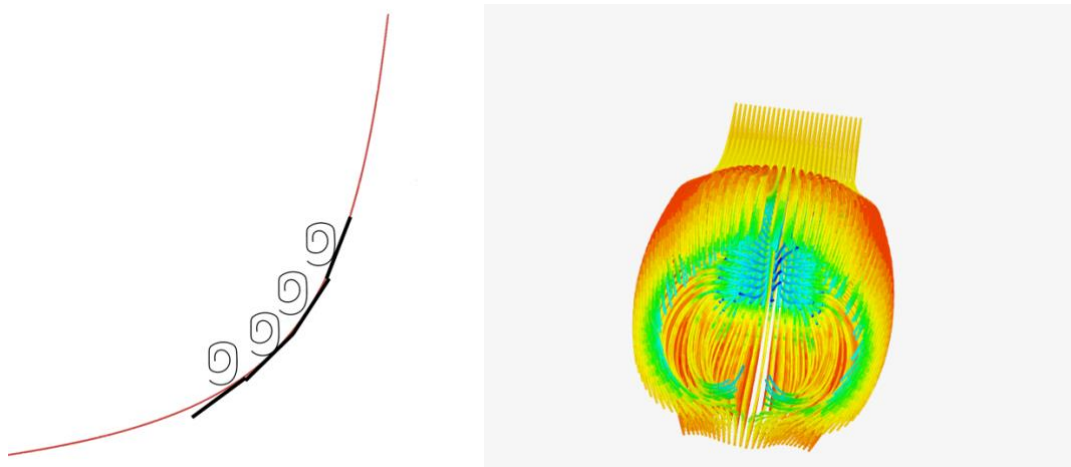


Figure 10. Splitting of the air flow caused by the number of vertices at a dilation factor of 1 (left side) (Own diagram) and the pressure difference created (right side) was made using SimScale.

The limited range of wind speeds, due to the limitation of equipment, does not allow me to draw completely accurate trends between the coefficient of downforce and the wind speeds. Although extrapolating the linear relationship suggests a negative downforce coefficient outside the tested range, within the measured range, the linear model provides the best fit. Additionally, these wind speeds were not particularly close to actual speeds that F1 cars face, once again due to the lack of equipment that can produce wind speeds upwards of 30ms^{-1} . The use of a turbulent source of air may have reduced the sensitivity to

localise velocity variation, making the air more uniform but resulting in unpredictability in how the custom modelled wing interacts with the air flow.

Another limitation of this investigation was that it did not completely replicate how an F1 car moves through the wind. While this is a common method used in aerodynamic testing, it does not perfectly reflect real-world conditions. A possible future improvement would be mounting the rear wing on a moving car to better simulate air flow. This would require a data-logging scale, as reading the scale during motion is impractical and unsafe. This setup, shown in Figure 11, was not feasible due to resource limitations.

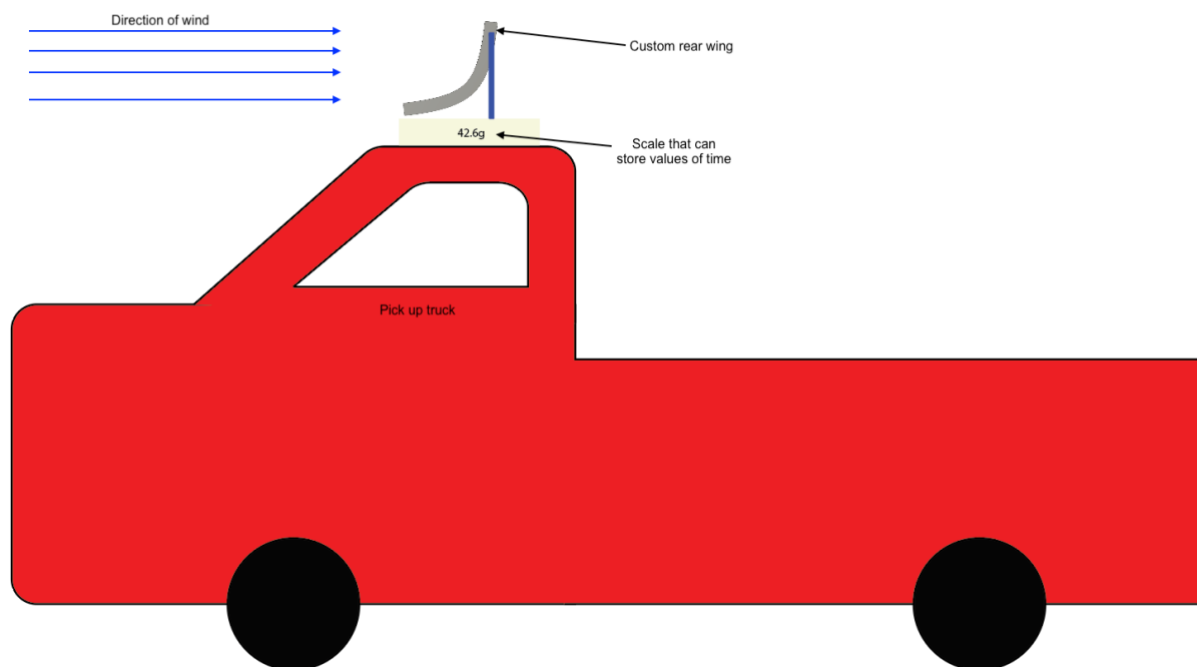


Figure 11. Future set-up that would replicate how an F1 rear wing moves through the wind (Own Diagram).

Overall, this investigation identified two relationships: one between the dilation factor and the coefficient of downforce and the other between the coefficient of downforce and the wind speed.

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